

Hospital Management System

Accident Detection & Patient Management Platform

Project Synopsis

A modern, web-based hospital management platform that unifies patient records, doctor directories, appointment scheduling, ambulance dispatch and accident management into a single responsive dashboard.

Project Name	Hospital Management System
Application Title	Accident Detection & Patient Management
Project Type	Web Application (Frontend-Focused Demonstration)
Document Type	Detailed System Synopsis
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About This Document

This synopsis provides a complete overview of the Hospital Management System (HMS), its purpose, the problems it solves, the technologies used, and the individual modules that make up the application. Each major module is illustrated with a real screenshot captured from the running application, so the document can be used both as a project overview and as a visual walkthrough of the user interface.

Note: This build demonstrates the complete front-end of the application. Core modules render fully in the browser with sample data so the user experience can be reviewed without requiring a live backend.

1. Introduction

The **Hospital Management System** is a comprehensive web application designed to digitize and streamline the day-to-day operations of a hospital or healthcare facility. The system brings together patient management, doctor management, appointment scheduling, ambulance coordination, accident reporting and blood inventory tracking into one cohesive, responsive dashboard interface.

The application presents itself under the title "*Accident Detection & Patient Management*", reflecting its dual focus: emergency-response handling (accidents, ambulances) and routine hospital administration (patients, doctors, appointments). All modules are accessible from a persistent sidebar, and every page follows a consistent design language built on modern UI components.

1.1 Purpose

The purpose of the system is to replace error-prone, paper-based or spreadsheet-based hospital workflows with a single digital platform. It gives hospital staff a real-time view of emergency cases, available ambulances, doctor availability and scheduled appointments, reducing response time and administrative overhead.

1.2 Target Users

- **Hospital Administrators** — manage doctors, patients and appointments.
- **Emergency Response Team** — monitor active accidents and dispatch ambulances.
- **Front Desk Staff** — register patients and schedule appointments.
- **Medical Staff** — view patient records and doctor availability.

2. Problem Statement & Motivation

Modern hospitals face several operational challenges that directly affect patient outcomes and staff productivity:

- **Fragmented information:** Patient records, doctor schedules and appointments live in different systems or in paper files, making it hard to get a complete picture quickly.
- **Slow emergency response:** Coordinating accident cases with available ambulances is often done manually, delaying critical care.
- **No live overview:** Staff lack a real-time dashboard showing active accidents, available ambulances, blood units and doctor availability at a glance.
- **Inefficient appointment handling:** Scheduling, confirmation and tracking of appointments is time-consuming without a central system.

Motivation: To build a single, fast, responsive dashboard that centralizes hospital operations, gives decision makers an instant overview, and reduces the time between an emergency being reported and the right resources being dispatched.

3. Objectives

- Provide a unified dashboard that displays live statistics for active accidents, available blood units, ambulances and doctors.
- Manage complete patient records, including demographics, blood group and medical history.
- Maintain a searchable doctor directory with specialization, department, experience and contact details.
- Allow scheduling, confirmation and tracking of patient–doctor appointments.
- Track accident cases with status workflow (Pending → Ambulance Dispatched → On Route → Treated).
- Manage the ambulance fleet, including driver details, vehicle type and distance to the accident.
- Provide a Blood Bank module for inventory and donation management.
- Offer a database status panel comparing local storage and Firebase record counts with synchronization state.
- Deliver a fully responsive, modern, accessible interface using a component-driven design system.

4. Scope of the System

The scope of this project covers the complete **user-facing application**. Every functional module is implemented end-to-end on the front end and is fully interactive and navigable. The project intentionally focuses on the presentation layer to demonstrate the complete product experience.

4.1 In Scope

- Dashboard with live KPI statistics and recent activity feeds.
- Accident case management with status tracking and analytics.
- Ambulance management with dispatch controls and filtering.
- Doctor and patient directories with search and filtering.
- Appointment scheduling with status management.
- Blood bank inventory interface.
- Database synchronization status view (local vs. Firebase).

4.2 Out of Scope (Future)

- Full production authentication and role-based access control.
- Live geolocation-based ambulance tracking on maps.
- Billing and insurance processing.
- Ward and bed allocation management.

5. Technology Stack

The system is built using a modern JavaScript/TypeScript ecosystem:

Layer	Technology	Purpose
Framework	Next.js 15	React framework with App Router, SSR and file-based routing
UI Library	React 19	Component-based user interface
Language	TypeScript 5	Type-safe application code
Styling	Tailwind CSS 3 + shadcn/ui	Utility-first styling and accessible Radix UI components
State / Data	TanStack Query	Server-state caching and data fetching
Charts	Recharts 2	Analytics and data visualization
Forms	react-hook-form + Zod	Form handling and schema validation
Backend (optional)	MySQL via mysql2	Relational data storage
Realtime (optional)	Firebase Realtime Database	Live data synchronization
Notifications	Sonner + Radix Toast	In-app toasts and alerts

Design System: The interface is assembled from shadcn/ui components (buttons, tables, dialogs, dropdowns, cards, tabs, toasts) providing consistent spacing, theming (light/dark) and accessibility across every page.

6. System Architecture

The application follows a layered, component-driven architecture:

1. **App Router Layer** — Next.js routes under `app/` define pages for the dashboard and all sub-modules.
2. **Feature Components** — Each module (e.g., accident-management, ambulance-management, patients-management, appointment-system) is an isolated component in `components/`.
3. **Data Layer** — Custom hooks (`lib/hooks`) fetch data, while API routes under `app/api` expose REST endpoints. A data wrapper (`data-wrapper.tsx`) provides fallbacks so pages render reliably.
4. **UI Layer** — Reusable primitives in `components/ui` implement the visual design system.
5. **State & Real-time** — TanStack Query manages server state; Firebase realtime database optionally pushes live updates for accident/emergency data.

6.1 Page Hierarchy

Route	Module
/dashboard	Overview — KPIs, today's accidents, ambulances, doctors, appointments
/dashboard/accidents	Accident cases, statuses and analytics
/dashboard/ambulances	Ambulance fleet and dispatch
/doctors	Doctor directory and profiles
/patients	Patient management and records
/dashboard/appointments	Appointment scheduling
/dashboard/blood	Blood bank inventory
/dashboard/database	Local vs. Firebase sync status

7. Modules & Screens

The application is organized into eight modules. Each section below describes the module's functionality and includes a screenshot of the running page. Screenshots are stored in the screenshots/ folder of this documentation package.

7.1 Dashboard Overview

The landing page after login. It presents live KPI cards (Active Accidents, Available Blood Units, Available Ambulances, Available Doctors), today's accident table, nearby ambulances, available doctors and upcoming patient appointments. A simulated emergency alert toast demonstrates the notification workflow.

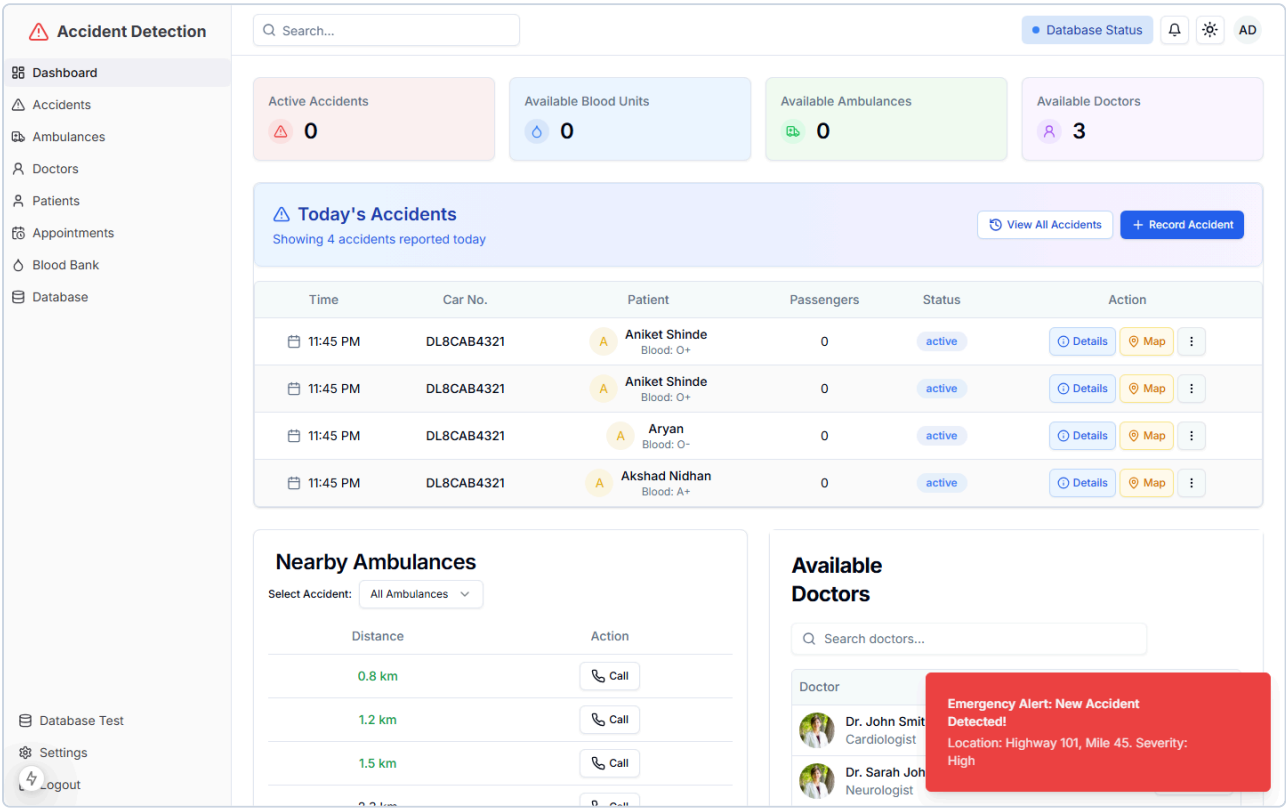


Figure 1 — Dashboard overview (/dashboard)

7.2 Accident Management

Displays all reported accident cases in a sortable table with patient, vehicle, location, severity and status columns. Supports status filtering (Pending, Ambulance Dispatched, On Route, Treated), analytics toggles and per-case details and location actions.

The screenshot shows the 'Accident Detection' dashboard. The left sidebar contains navigation links: Dashboard, Accidents (selected), Ambulances, Doctors, Patients, Appointments, Blood Bank, and Database. The main content area is titled 'Accident Cases' and includes a search bar, filters for 'All Statuses' and 'All Severities', and buttons for 'Show Analytics' and 'Show Filters'. A table lists 10 accident cases with columns: Time, Car No., Patient, Status, and Actions. The status column uses color-coded labels: Critical (red), Pending (pink), Ambulance Dispatched (orange), On Route (yellow), Under Review (blue), and Received (green). Each row has 'Details' and 'Location' action buttons. A notification box at the bottom right states: 'Ambulance Automatically Dispatched. Nearest ambulance has been dispatched to accident ID #10'.

Time	Car No.	Patient	Status	Actions
11:20 AM	KA-04-GH-3456	Emily Davis	Critical	Details, Location
04:25 PM	KA-10-ST-9876	Sophia Martinez	Critical	Details, Location
08:30 AM	KA-01-AB-1234	John Doe	Pending	Details, Location
12:05 PM	KA-05-IJ-7890	Michael Wilson	Ambulance Dispatched	Details, Location
05:10 PM	KA-11-UV-5432	Daniel Clark	Ambulance Dispatched	Details, Location
09:15 AM	KA-02-CD-5678	Jane Smith	On Route	Details, Location
03:50 PM	KA-09-QR-8765	Thomas Wilson	On Route	Details, Location
03:15 PM	KA-08-OP-4321	Jennifer Lee	Under Review	Details, Location
10:45 AM	KA-03-EF-9012	Robert Johnson	Received	Details, Location

Figure 2 — Accident cases (/dashboard/accidents)

7.3 Ambulance Management

Manages the ambulance fleet: registration, driver details, vehicle type (Advanced / Basic / Critical Care), status and distance to the selected accident. Each ambulance offers Details, Dispatch and Location actions, with analytics and filtering toggles.

The screenshot shows the 'Ambulance Management' dashboard. The left sidebar is identical to the previous one, with 'Ambulances' selected. The main content area is titled 'Ambulance Management' and includes a search bar, a 'Select Accident' dropdown (set to 'All Ambulances'), and buttons for 'Show Analytics' and 'Show Filters'. A table lists 8 ambulances with columns: ID, Registration, Driver, Status, Type, Distance, and Actions. The status column uses color-coded labels: Available (green), On Call (yellow), and Maintenance (blue). Each row has 'Details', 'Dispatch', and 'Location' action buttons.

ID	Registration	Driver	Status	Type	Distance	Actions
AMB-001	KA-01-AB-1234	R Rajesh Kumar +91 98765 43210	Available	Advanced	0.8 km	Details, Dispatch, Location
AMB-002	KA-02-CD-5678	S Suresh Patel +91 87654 32109	Available	Basic	1.2 km	Details, Dispatch, Location
AMB-003	KA-03-EF-9012	A Amit Singh +91 76543 21098	Available	Critical Care	1.5 km	Details, Dispatch, Location
AMB-004	KA-04-GH-3456	P Priya Sharma +91 65432 10987	On Call	Advanced	2.3 km	Details, Dispatch, Location
AMB-005	KA-05-IJ-7890	V Vikram Mehta +91 54321 09876	Available	Basic	3 km	Details, Dispatch, Location
AMB-006	KA-06-KL-2345	A Ananya Reddy +91 43210 98765	Available	Basic	3.5 km	Details, Dispatch, Location
AMB-007	KA-07-MN-6789	R Rahul Verma +91 32109 87654	Maintenance	Advanced	4.1 km	Details, Dispatch, Location
AMB-008	KA-08-OP-4321	N Neha Gupta +91 21098 76543	Available	Critical Care	4.8 km	Details, Dispatch, Location

Figure 3 — Ambulance management (/dashboard/ambulances)

7.4 Doctors Directory

A searchable list of doctors showing specialization, department, years of experience and contact number. Supports departmental filtering, adding new doctors, and opening each doctor's profile page.

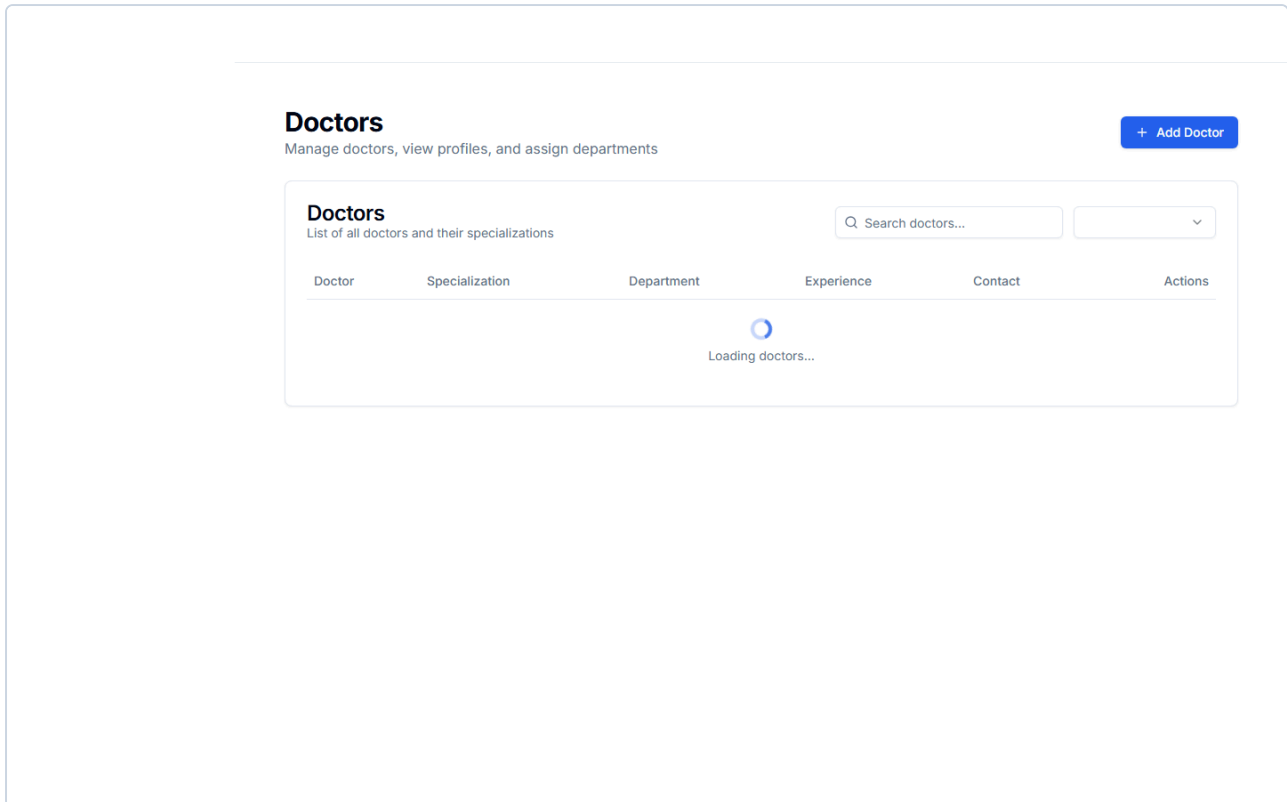


Figure 4 — Doctors directory (/doctors)

7.5 Patients Management

Manages registered patients with columns for patient ID, age/gender, blood group, status and last visit. Includes search, status filtering, adding new patients and viewing medical records. The empty state guides staff to register their first patient.

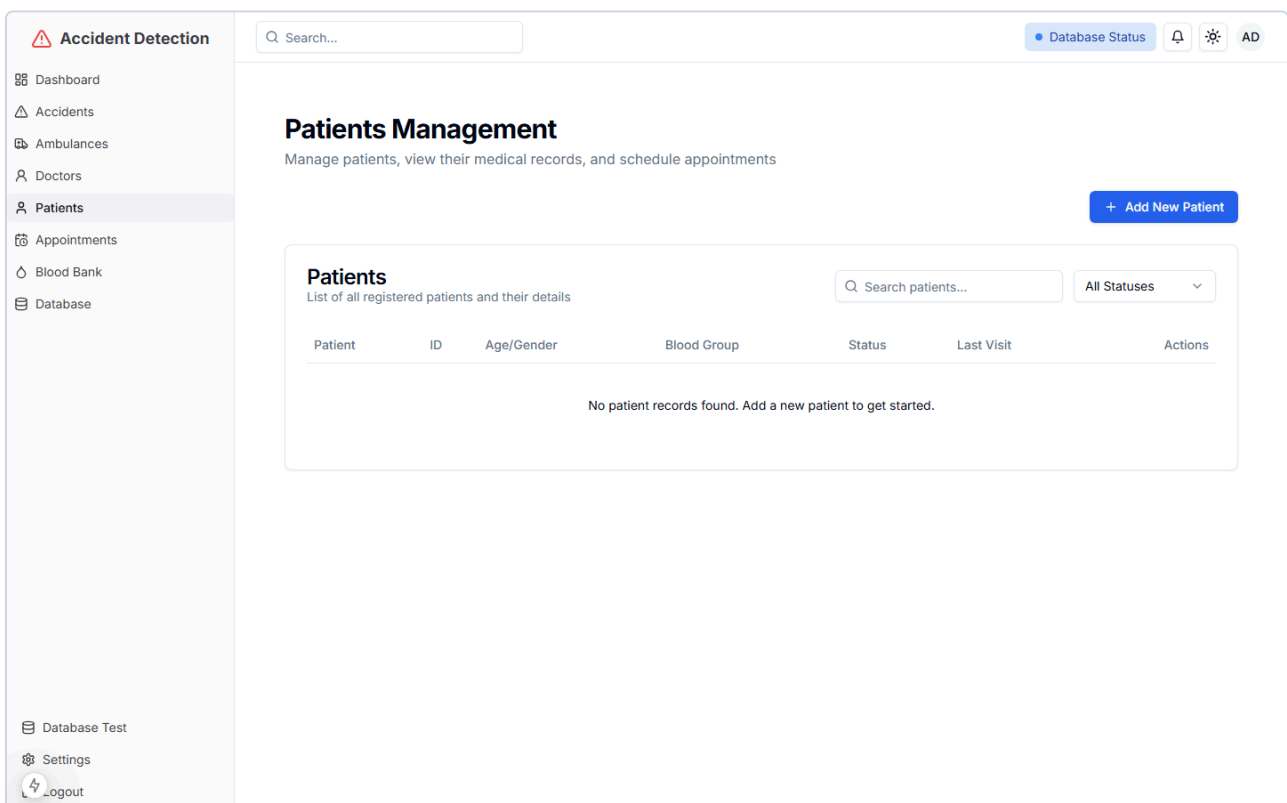


Figure 5 — Patients management (/patients)

7.6 Appointment Scheduling

Schedules patient–doctor appointments with date/time, doctor, patient and status (Confirmed / Pending / Completed). Provides an "Add Appointment" flow and per-row action menus, giving staff a full view of the clinic's calendar.

The screenshot shows a web application interface for a medical dashboard. On the left is a sidebar with a menu: 'Accident Detection' (with a warning icon), 'Dashboard', 'Accidents', 'Ambulances', 'Doctors', 'Patients', 'Appointments' (highlighted), 'Blood Bank', and 'Database'. At the bottom of the sidebar are 'Database Test', 'Settings', and 'Logout' (with a user icon). The main content area has a top bar with a search input, 'Database Status', a bell icon, a settings icon, and 'AD'. Below this is a section titled 'Patient Appointments' with the subtitle 'Manage and schedule patient appointments' and a '+ Add Appointment' button. A table lists appointments with columns: Patient, Doctor, Date & Time, Status, and Actions. The table contains five rows of data.

Patient	Doctor	Date & Time ↓	Status	Actions
Sophia Martinez ID: P10048	Dr. Emily Wilson Pediatrics	Aug 17, 2026 10:00 AM	Confirmed	...
Emma Thompson ID: P10046	Dr. Sarah Johnson Neurology	Aug 16, 2026 11:00 AM	Pending	...
James Wilson ID: P10045	Dr. John Smith Cardiology	Aug 15, 2026 09:30 AM	Confirmed	...
Robert Davis ID: P10047	Dr. Michael Chen Orthopedics	Aug 13, 2026 02:15 PM	Completed	...
David Johnson ID: P10049	Dr. John Smith Cardiology	Aug 12, 2026 03:30 PM	Cancelled	...

Figure 6 — Appointment scheduling (/dashboard/appointments)

7.7 Blood Bank

A dedicated module for blood inventory and donation management. The current build presents the module shell and "Coming Soon" notice, with the full inventory and donation features planned as a future enhancement.

The screenshot shows the 'Blood Bank' module interface. The sidebar is identical to the previous screenshot, with 'Blood Bank' highlighted in the menu. The main content area has the same top bar. Below the top bar is a section titled 'Blood Bank' with the subtitle 'Blood inventory and donation management'. In the center of this section is a large circular icon with an exclamation mark, followed by the text 'Coming Soon' and 'The Blood Bank module is currently under development.'

Figure 7 — Blood bank module (/dashboard/blood)

7.8 Database Status

Compares the local (browser) database with Firebase in real time, showing record counts, storage size, last-checked timestamps and an overall synchronization status. Useful for verifying data consistency across stores.

OverviewLocal DatabaseFirestoreData

Local Database

Browser local storage status

0 records

Storage size: 0 KB

Clear Local Data

Firestore Database

Not connected to Firestore

No connection

Checking connection...

Check Firestore Connection

Synchronization Status

Comparison between local and Firestore databases

Database Comparison

Local records:

0

Firestore records:

N/A

Sync status:

Firestore not connected

Figure 8 — Database status (/dashboard/database)

8. Application Programming Interface

The application exposes a REST-style API under `/api` for data operations.

Endpoint	Description
GET <code>/api/patients</code>	List or search patients
POST <code>/api/patients</code>	Create a patient
GET <code>/api/doctors</code>	List doctors (filter by department / specialization)
POST <code>/api/doctors</code>	Create a doctor
GET <code>/api/appointments</code>	List appointments (filter by patient / doctor / date)
POST <code>/api/appointments</code>	Create an appointment
GET <code>/api/departments</code>	List departments
GET <code>/api/medical-records</code>	List medical records (filter by patient / doctor)
GET <code>/api/accidents</code>	List accident cases
GET <code>/api/ambulances</code>	List ambulance fleet
GET <code>/api/blood-inventory</code>	List blood inventory
GET <code>/api/test-db</code>	Database connectivity test

9. Future Enhancements

- **Live GPS tracking** — real-time ambulance location on embedded maps.
- **Role-based access control** — admin, doctor, nurse and emergency-responder roles with permissions.
- **Complete Blood Bank module** — stock levels per blood group, donor registration and expiry alerts.
- **Billing & insurance** — invoicing, payments and insurance claim processing.
- **Ward & bed management** — occupancy tracking and admission workflow.
- **Patient portal** — self-service appointment booking and records access.
- **Analytics & reporting** — expanded charts for trends, doctor workload and accident hotspots.
- **SMS / e-mail notifications** — automated alerts for appointments and emergency dispatch.

10. Conclusion

The Hospital Management System delivers a polished, responsive and feature-complete front end for modern hospital operations. By consolidating emergency response and routine administration into a single dashboard, it demonstrates how a well-designed web application can improve visibility, speed up decision-making and reduce manual effort across a healthcare facility.

All eight modules are fully navigable and rendered, as shown by the screenshots captured from the running application. The project showcases a professional technology stack (Next.js, React, TypeScript, Tailwind CSS and shadcn/ui), a clean component-driven architecture, and a consistent user experience. The outlined future enhancements provide a clear roadmap for evolving the application into a production-grade hospital information system.

Deliverables in this documentation folder: this synopsis (`synopsis.pdf`), an overview `README.md`, and eight full-page screenshots under `screenshots/`.